Serialize and Deserialize Binary Tree (hard)

Problem Statement

Given a binary tree, your task is to create two functions.

one for serializing the tree into a string format and another for deserializing the string back into the tree.

The serialized string should retain all the tree nodes and their connections, allowing for reconstruction without any loss of data.

Examples

1. **Example 1:**
   - **Input:** `[1,2,3,null,null,4,5]`
   - **Expected Output:** `[1,2,3,null,null,4,5]`
   - **Justification:** The tree has the structure:

```
  1
 / \
2   3
   / \
  4   5
```

   When serialized and then deserialized, it should retain the exact same structure.

2. **Example 2:**
   - **Input:** `[1,null,2,3]`
   - **Expected Output:** `[1,null,2,3]`
   - **Justification:** The tree has the structure:

```
  1
   \
    2
   /
  3
```

   When serialized and then deserialized, it should retain the exact same structure.

3. **Example 3:**
   - **Input:** `[5,4,7,3,null,null,null,2]`
   - **Expected Output:** `[5,4,7,3,null,null,null,2]`
   - **Justification:** The tree has the structure:

```
      5
    /   \
   4     7
  /
 3
/
2
```

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

public class Solution {
//    public static class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;
//     TreeNode(int x) { val = x; }
//     }

    // Encodes a tree to a single string using a pre-order traversal.
    public String serialize(TreeNode root) {
        StringBuilder sb = new StringBuilder();
        // ToDo: Write Your Code Here.
        return sb.toString();
    }

    // Decodes your encoded data to tree.
    public TreeNode deserialize(String data) {
        TreeNode node = new TreeNode(0);
        // ToDo: Write Your Code Here.
```

Test Case | Run Code Result | Stdout

[1,2,3,null,null,4,5]

Use Example Testcases | Run | Submit